

Saha Malaki

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[Saha Malaki](#) • [Google Scholar](#) • [Website](#)

Professional Employment

Postdoctoral Fellow

Rotman School of Management, University of Toronto

Sep 2026–Present

Remote from Seattle, WA, USA

Supervisor: Olga Bountali.

- Developing queueing-game and optimization models of recurrent service access, strategic re-entry, and reassessment, motivated by dialysis for uninsured patients.

Postdoctoral Associate

Department of Statistical and Actuarial Sciences, Western University

May 2024–Oct 2025

London, ON, Canada

Supervisors: Felipe Rodrigues, Greg Zaric, and Salar Ghamat.

- Developed queueing-game and Python simulation models of autism-service access and funding policies; resulting manuscript: invited to resubmit at *Production and Operations Management*; resubmission in preparation.
- Developed a game-theoretic model of post-patent pharmaceutical pricing and market access; resulting working paper.
- Contributed to research design, methodology, and writing for a 2025 SSHRC Insight Grant application.

Graduate Research Assistant (Doctoral Research)

Bayes Business School (formerly Cass), City, University of London

Sep 2018–Jan 2023

London, UK

Developed stochastic optimization and Markov decision process models for healthcare workforce planning and calibrated inpatient-flow simulations using hospital data in Python and R.

Graduate Research Assistant (Master's Research)

University of Alberta

Sep 2014–Sep 2016

Edmonton, AB, Canada

Developed mixed-integer programming models for block-cave production scheduling under grade uncertainty using MATLAB, CPLEX, and geostatistical simulation.

Research Interests

Methods: stochastic and dynamic optimization; queueing theory; Markov decision processes; discrete-event simulation; game theory; mathematical programming; data analytics and learning-based decision methods.

Applications: healthcare operations and analytics; public-service systems; workforce and capacity planning; service access and incentives; pharmaceutical markets.

Education

Ph.D., Operations Management

Bayes Business School (formerly Cass), City, University of London

Sep 2017–Jan 2023

London, UK

Dissertation: *Optimal Recruitment of Temporary and Permanent Healthcare Workers in Highly Uncertain Environments*.

Supervisors: Navid Izady, Lilian M. de Menezes, and Oben Ceryan.

M.Sc., Engineering

University of Alberta

Sep 2014–Sep 2016

Edmonton, AB, Canada

Thesis: *Block-Cave Extraction Level and Production Scheduling Optimization under Grade Uncertainty*.

Supervisor: Yashar Pourrahimian.

B.Sc., Industrial Engineering

K. N. Toosi University of Technology

Sep 2008–Sep 2012

Tehran, Iran

Publications and Research Pipeline

Peer-Reviewed Journal Articles

Saha Malaki, Navid Izady, Oben Ceryan, and Lilian M. de Menezes. “Optimal Recruitment of Temporary and Permanent Healthcare Workers in Highly Uncertain Environments: A Long-Term Scenario.” *European Journal of Operational Research*, 336(1):208–223, 2027.

Saha Malaki, Navid Izady, and Lilian M. de Menezes. “A Framework for Optimal Recruitment of Temporary and Permanent Healthcare Workers in Highly Uncertain Environments.” *European Journal of Operational Research*, 308(2):768–781, 2023.

Manuscript for Resubmission

Saha Malaki, Felipe Rodrigues, Salar Ghamat, Greg Zaric, and Camila de Souza. “Optimal Contract Policies for Improving Access to Child and Youth Public Funding: The Case of Ontario Autism Program.” Invited to Resubmit at *Production and Operations Management*; resubmission in preparation.

Working Papers

“Strategic Re-entry in Recurrent Service Systems: When to Return and How to Manage Reassessment.” University of Toronto.

“Pricing Competition and Market Access in the Post-Patent Pharmaceutical Market: A Game-Theoretic Approach.” Western University.

Peer-Reviewed Conference Proceeding

Saha Malaki, Firouz Khodayari, Yashar Pourrahimian, and Wei Victor Liu. “An Application of Mathematical Programming and Sequential Gaussian Simulation for Block Cave Production Scheduling.” *Proceedings of the First International Conference on Underground Mining Technology*, 323–337, 2017.

Teaching Experience

SUBSTITUTE LECTURER

Quantitative Methods and Analytics

Bayes Business School

2018–2022

London, UK

Delivered several class sessions; worked through examples and facilitated discussion.

CLASS SESSION

Support Vector Machines, Workforce Analytics

Imperial College Business School

Winter 2020

London, UK

Developed instructional content and delivered one graduate class session.

GRADUATE TEACHING ASSISTANT

Supply Chain Management Analytics, University of London — tutorials and exam marking; Spring 2022.

Operations Management, Bayes Business School (online) — tutorials and project assessment; Spring 2021.

Data Analytics for Management and Data Analytics for Business, London Business School (online) — tutorials; Fall 2020.

Business Forecasting and Data Analytics, University of London (online) — tutorials and project assessment; Fall 2020.

Workforce Analytics, Imperial College Business School — tutorials, office hours, and report/project assessment; Winter–Spring 2020.

UNDERGRADUATE TEACHING ASSISTANT

Quantitative Methods and Analytics, Bayes Business School — tutorials; Fall 2018–2021.

Operations and Supply Chain Management, Bayes Business School — tutorials; Winter 2020–2021.

Mechanics of Deformable Bodies I, Engineering Mechanics, Plane Surveying, and Engineering Drawing and Computer Graphics, University of Alberta — tutorials, office hours, assessment, and project/fieldwork support; 2015–2016.

PEDAGOGICAL CREDENTIAL AND EVIDENCE

Associate Fellow of the Higher Education Academy (AFHEA), City, University of London, 2019.

Individual Workforce Analytics teaching-assistant composite: 3.72/5. Quantitative Methods tutorial-teaching result: 3.86/5 (module/tutorial-team item, 126 responses). Student comments named Saha for clear explanations and responsiveness.

Teaching Areas

Operations Management; Supply Chain Analytics; Production Planning and Control; Quantitative Supply Chain Methods; Business Statistics; Forecasting; Simulation; Optimization.

Methods and Technical Skills

OR and simulation: stochastic optimization; queueing; Markov decision processes; dynamic programming; game theory; mixed-integer and linear programming; discrete-event simulation.

Analytics and machine learning: regression, classification, clustering, forecasting, supervised and unsupervised learning, reinforcement learning, neural networks, and interpretable models.

Programming and tools: Python, R, SQL, MATLAB, CPLEX, Mathematica, Tableau, Excel, and L^AT_EX; Pandas, NumPy, SciPy, scikit-learn, TensorFlow, Matplotlib, and ggplot.

Mentoring and Academic Contributions

Master's student mentoring, during Ph.D.: mentored two students developing Python discrete-event simulation models of healthcare service systems; supported scoping, model development, debugging, validation, and interpretation.

Research communication: presented work in Bayes Business School's Operations Management Seminar Series (Oct 2020) and City, University of London's Economics Internal Seminar Series (May 2020).

Conference Presentations

INFORMS Annual Meeting, Seattle, WA, USA, Oct 2024. "Optimal Contract Policies for Improving Access to Child and Youth Public Funding: The Case of Ontario Autism Program."

Canadian Operational Research Society Conference, London, ON, Canada, Jun 2024. "Long-Term Blended Workforce Planning in Healthcare."

MSOM Conference, Munich, Germany, Jun 2022. "A Framework for Optimal Recruitment of Temporary and Permanent Healthcare Workers in Uncertain Environments."

INFORMS Healthcare Conference, online, Jul 2021. "Long-Term Blended Workforce Planning in Healthcare."

POMS Conference, Brighton, UK, Sep 2019. "Optimal Balance Between Temporary and Permanent Staff in Healthcare Systems."

Selected Awards

Bayes Business School Doctoral Studentship, 2017–2022 (tuition and stipend).

Prairie Mines & Royalty Ltd Graduate Scholarship in Environmental Engineering, University of Alberta, 2015 (CAD 10,000).

University of Alberta Graduate Scholarship, 2014–2016 (CAD 23,000 per year).

EURO Ph.D. School Registration Scholarship, Universidad de Sevilla, Jul 2020 (EUR 1,300).

References

Greg Zaric | Professor, Management Science; former postdoctoral supervisor
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Olga Bountali | Assistant Professor; current postdoctoral supervisor
Rotman School of Management, University of Toronto
105 St. George Street, Toronto, ON M5S 3E6, Canada
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Felipe Rodrigues | Associate Professor; former postdoctoral supervisor
King's University College, Western University
266 Epworth Avenue, London, ON N6A 2M3, Canada
frodrig7@uwo.ca • +1 (519) 518-0949

Navid Izady | Reader in Operations & Supply Chain; Ph.D. supervisor
Bayes Business School, City St George's, University of London
106 Bunhill Row, London EC1Y 8TZ, UK
navid.izady@citystgeorges.ac.uk • +44 (20) 7040 0290

Lilian M. de Menezes | Professor of Decision Sciences; Ph.D. supervisor
Bayes Business School, City St George's, University of London
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